

Potential Outcomes, Identification Logic, and Selection-On-Observables

Empirical Methods — Week 1

Teaching deck draft

Week 1 question

- What does it mean to identify a causal effect?
- What is observed, what is missing, and what assumption links them?
- When can rich observables support a credible comparison?

The lecture uses LaLonde, Dehejia–Wahba, Rosenbaum–Rubin, and sensitivity-style work as the paper spine.

$$Y_i = D_i Y_i(1) + (1 - D_i) Y_i(0)$$

$$\tau_i = Y_i(1) - Y_i(0)$$

- Treated units reveal $Y_i(1)$ and hide $Y_i(0)$.
- Untreated units reveal $Y_i(0)$ and hide $Y_i(1)$.
- Identification is an argument for replacing the missing potential outcome with a credible comparison.

$$ATE = \mathbb{E}[Y(1) - Y(0)]$$

$$ATT = \mathbb{E}[Y(1) - Y(0) \mid D = 1]$$

$$ATC = \mathbb{E}[Y(1) - Y(0) \mid D = 0]$$

- ATE: average effect for the population represented by the sample.
- ATT: effect for units who actually received treatment.
- ATC: effect for units who are currently untreated.
- Support restrictions and weights can quietly change the parameter.

Selection bias decomposition

$$\mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0] = \underbrace{\mathbb{E}[Y(1) - Y(0) \mid D = 1]}_{ATT} + \underbrace{\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]}_{\text{selection bias}}$$

- The raw comparison is causal only if untreated potential outcomes are comparable.
- LaLonde shows how badly this can fail with the wrong nonexperimental comparison group.
- The comparison group is the design.

Conditional independence and overlap

$$\{Y_i(1), Y_i(0)\} \perp D_i \mid X_i$$

$$0 < \mathbb{P}(D_i = 1 \mid X_i = x) < 1 \quad \text{for relevant } x$$

- Conditional independence: observables capture the assignment channels that matter for potential outcomes.
- Overlap: treated and untreated units coexist in the relevant covariate region.
- The propensity score balances observables under the identifying assumption; it does not create the assumption.

Selection-on-observables is strongest when

- treatment assignment is institutionally understood;
- pre-treatment outcomes and histories are observed;
- covariates are measured before treatment;
- untreated comparisons come from the same economic environment;
- common support is strong;
- the main selection margins are observed rather than private or latent.

Regression, matching, weighting

Approach	Design idea	Main risk
Regression adjustment	Model $\mathbb{E}[Y \mid D, X]$ and compare adjusted outcomes	Functional-form extrapolation
Matching	Compare treated units to similar untreated units	Bad matches or support loss
Weighting	Reweight by $e(X) = \Pr(D = 1 \mid X)$	Extreme weights when overlap is weak

All three operationalize the same identifying claim: conditional comparability on observed pre-treatment information.

$$ATE = \mathbb{E} \left[\frac{DY}{e(X)} - \frac{(1-D)Y}{1-e(X)} \right]$$

$$\mathbb{E} \left[\hat{m}_1(X) - \hat{m}_0(X) + \frac{D(Y - \hat{m}_1(X))}{\hat{e}(X)} - \frac{(1-D)(Y - \hat{m}_0(X))}{1 - \hat{e}(X)} \right]$$

- Weighting uses the treatment model to reconstruct a target population.
- Doubly robust estimators combine treatment and outcome models.
- Neither relaxes the need for a credible observables-based assignment story.

Papers as design lessons

Paper role	Design lesson
LaLonde	Wrong comparison groups can miss the experimental benchmark badly.
Dehejia–Wahba	Matching can be credible when covariates, sample, and support are chosen deliberately.
Rosenbaum–Rubin	Propensity scores are balancing scores under conditional independence.
Altonji–Elder–Taber / Oster	Sensitivity work asks how strong omitted selection would need to be.

- Balance: standardized differences before and after adjustment.
- Overlap: propensity-score support, trimming, and tails.
- Matching quality: exact variables, calipers, and substantively plausible pairs.
- Weighting stability: large weights and influential observations.
- Specification discipline: pre-treatment covariates and transparent model families.
- Sensitivity: omitted-variable robustness, placebo outcomes, negative controls.

Reproduce

- Synthetic LaLonde-style training data
- Naive, regression, matching, IPW, doubly robust estimates

Diagnose

- Balance
- Overlap
- Weight stability
- Estimand drift

Transfer

- Workplace training take-up
- Target parameter
- Selection story
- Omitted-variable threat

Takeaways

- Potential outcomes force the missing counterfactual into view.
- Selection bias is an economic assignment problem, not a regression nuisance.
- Selection-on-observables is credible only when observables capture the relevant selection story and support is real.
- Diagnostics and sensitivity analysis are part of the design, not an appendix ritual.
- Next week: experiments change the assignment mechanism, but not the need for estimand discipline.